ACE
APPS

Vocabulary Builder

Full Feature Analysis

A technical & competitive deep-dive

Science-Backed | B2B-Ready | Vietnam-Focused

14	5	12
Slides	Competitors	Feature categories

The SM-2 Spaced Repetition Engine

Ease Factor

Starting value

2.5

Multiplier applied to the review interval after each successful recall. Increases with strong recall, decreases with poor recall. Range: 1.3 – 2.5+.

Review Interval

Until next review

Days

Days before the word is shown again. Starts at 1 day and grows exponentially: 1 !’ 3 !’ 7 !’ 14 !’ 30 !’ 60+ as recall improves.

Review Count

Per word

N+

Number of times a word has been reviewed. Combined with accuracy to determine mastery level promotion: Learning !’ Reviewing !’ Mastered.

Poor recall **resets the interval**; strong recall **extends it exponentially** — optimizing long-term retention while minimizing study time.

Day 1	Day 3	Day 7	Day 14	Day 30+
Review 1	Review 2	Review 3	Review 4	Review 5

Three Mastery Levels & Flashcard Mode

Learning
Default — every new word starts here.

Start

Reviewing
3+ reviews · "e70% accuracy

Progressing

Mastered
10+ reviews · "e90% accuracy · "e30-day interval

Complete

FLASHCARD MODE

Front of card

procurement

noun · /pr%'Ekj(¢Yrm%--çBó

Back of card (after flip)

The process of obtaining goods or services for a business.

"The procurement team reduced costs by 30%."

- Front/back review with manual flip
- Audio pronunciation from Dictionary API
- Example sentences on back of card
- SM-2 score recorded on every card flip

Quiz Mode & Daily Streak

Quiz Mode

Multiple-choice knowledge testing

Format	Multiple choice — 4 options per question
Session size	Up to 10 words per quiz session
Question type	Definition shown ⁉ pick the correct word
Scoring	Score + percentage displayed at end
SM-2 impact	Quiz results feed back into the spaced repetition engine
Distractor pool	Wrong options drawn from the user's own word library

Daily Streak

Consistency measurement & motivation

5 Day streak — keep going!

- Increments on consecutive practice days
- Resets if a day is missed
- Tracked per user account
- A proven motivational mechanic for habit-building

Word & Content Management

05

Manual Word Entry

Word
The target vocabulary word

Definition
Auto-fetched or manually entered

Example sentence
From dictionary API or your own

Custom tags
e.g. "finance", "Q3-review", "medical"

All words are editable and deletable at any time.

Search, Filter & Tag System

- **Live search**
  Searches across word text and definition in real time.
- **Filter by mastery**
  Show only Learning, Reviewing, or Mastered words.
- **Filter by tag**
  Narrow to a specific topic, project, or batch.
- **Custom tags per word**
  Attach one or more tags to any word at any time.
- **Batch tag on import**
  Apply a single tag to the entire CSV import batch.

Automatic Dictionary Lookup

Free Dictionary API

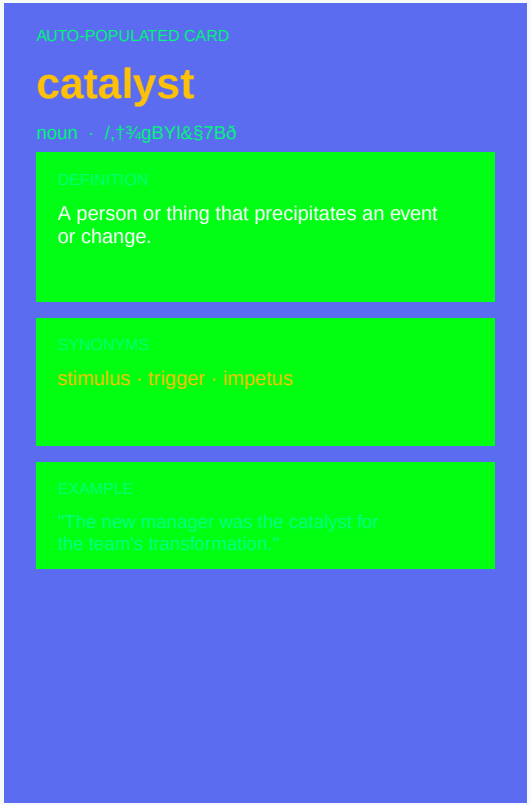
- Word definition
- Phonetic pronunciation (IPA)
- Audio pronunciation file
- Example sentences

Datamuse API

- Synonyms for the word
- Antonyms for the word
- Related word associations
- Triggered automatically on add

Zero friction word capture

Type a word !' definitions, examples, audio, synonyms, and antonyms populate automatically.



CSV Bulk Import — The Enterprise Accelerator

1

Auto-detect header row

Intelligently detects whether the first row is a header and skips it automatically.

2

Recognizes field variations

Accepts "word", "term", "vocab", "definition", "meaning" — flexible column naming.

3

Skips duplicate words

If a word already exists in the user's library, it is silently skipped — no duplicates created.

4

Skips invalid rows

Rows missing a word or definition are excluded and reported after import.

5

Preview before commit

Users see a full preview of parsed words with counts before confirming the import.

6

Apply tag to entire batch

One tag applied to all words in a single import — instant category organization.

This is the key feature for businesses loading industry-specific word lists.

Data Storage & Cross-Device Sync

Firebase Firestore — Cloud Vocabulary Storage

- **Real-time NoSQL database**
  Writes sync instantly across all devices without polling.
- **Synced across every device**
  Phone, tablet, laptop — one account, one library.
- **No manual backup needed**
  Firebase handles redundancy and durability automatically.
- **Data persists across reinstalls**
  Log back in to restore all words, tags, and mastery levels.
- **Per-UID security rules**
  Each user can only read and write their own data.

PostgreSQL — Subscriber Registry

- **Subscription status tracking**
  Active, trial, expired, or admin-granted — stored server-side.
- **Trial start and end dates**
  Precise timestamps for 7-day trial enforcement.
- **Admin manual activation**
  Admins can grant or extend access from the dashboard.
- **Payment event records**
  Paddle webhook events stored and linked to user accounts.
- **Survives server restarts**
  Unlike file-backed storage, data persists across redeployments.

Authentication & Security Architecture

09

Authentication — Firebase Auth v11

- **Email + Password signup/login**
  Standard credentials stored securely in Firebase Auth.
- **Password reset via email**
  Firebase sends a reset link — no server-side handling needed.
- **Anonymous sign-in**
  Users can subscribe without creating an account first.
- **Session persistence**
  Users remain logged in after closing and reopening the browser.

Security Layers

- **Helmet.js HTTP headers**
  Prevents XSS, clickjacking, and other common web attacks.
- **API rate limiting**
  Blocks brute-force and abuse on all API endpoints.
- **Paddle webhook signature**
  Every payment event is cryptographically verified before processing.
- **Firebase Security Rules per UID**
  Firestore enforces row-level security — users cannot access other users' data.
- **Admin password as env secret**
  Admin credentials stored as a server environment variable — never in the codebase.

Payments & Business Model

Pricing Structure

500,000 «

per 6 months (~$19 USD)

Base price	500,000 «
+ 10% VAT	50,000 «
Total charged	550,000 «

7-day free trial · Direct subscription (skip trial)

Paddle — International Payments

- Credit & debit cards (Visa, Mastercard)
- Apple Pay & Google Pay
- PayPal
- Global tax compliance (VAT handled by Paddle)
- Recurring billing with automatic renewal
- Paddle webhook — real-time payment events

VietQR — Local Bank Transfer

- User enters email in the paywall UI
- QR code revealed after email submission
- User transfers via any Vietnamese banking app
- Admin manually activates the account
- No Paddle subscription required for this path

Techcombank QR — the only local payment path for Vietnam in a vocabulary app at this price.

UI & User Experience

'&

Mobile-First Design

Vertical navigation, touch-friendly tap targets, and a native app feel — all in the browser. Optimised for 375px screens and up.

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Dark Mode

Full theme-aware toggle with a dark navy colour palette. Light mode also supported. User preference saved to localStorage.

'&

Responsive Layout

Scales cleanly from phone to tablet to desktop. The UI adapts its spacing, navigation, and card layout for each breakpoint.

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No App Store Required

Runs entirely in the browser — no iOS or Android download needed. Instant access from any device with a URL.

'&

Progressive Loading

Branded splash screen during auth check. Paywall state resolved before revealing the app. 3-second safety timeout fallback.

Competitive Comparison Table

Feature	Vocab Builder	Anki	Quizlet	Duolingo	Memrise
SM-2 Spaced Repetition	'	'	& þ	& þ	& þ
CSV Industry Word Import	'	& þ	& þ	'L	'L
Auto Dictionary + Audio	'	& þ	& þ	'	& þ
Synonyms / Antonyms	'	'L	'L	'L	'L
Cross-Device Cloud Sync	'	& þ	'	'	'
Works in Browser (no install)	'	'L	'	'	'
Dark Mode	'	'	'	'L	'L
Vietnamese Bank Payment	'	'L	'L	'L	'L
Admin Override / B2B Console	'	'L	& þ	'L	'L
Industry-Specific Focus	'	& þ	& þ	'L	'L
One-Time Bulk Team Setup	'	& þ	'L	'L	'L
Price (6 months)	~$19	Free/$25/yr	$35.99/yr	$79.99/yr	$59.99/yr

' Full support & þ Partial 'L Not available

The Core Competitive Arguments

vs Anki

- Anki requires manual deck setup, plugin installation, and desktop syncing.
- Vocabulary Builder runs in the browser — no install, no configuration.
- B2B CSV import and admin console have no equivalent in Anki.
- Anki has no Vietnamese payment option; VietQR is unique here.

vs Quizlet

- Quizlet's spaced repetition does not implement the full SM-2 algorithm.
- Quizlet charges $35.99/year vs 500,000 ₫/6 months (~$19 USD).
- No VietQR payment path and no admin override console in Quizlet.
- Our CSV import detects headers, skips duplicates, and previews before commit.

vs Duolingo / Babbel

- Duolingo and Babbel teach a fixed curriculum — no custom vocabulary.
- Neither supports CSV import or industry-specific word lists.
- Duolingo is $79.99/year — more than 4× the cost of this product.
- No B2B admin console or bulk team onboarding in either product.

vs Generic Vietnam Apps

- Most local alternatives use file-backed storage wiped on redeployment.
- No competitor offers both Paddle international billing AND VietQR.
- None provide an admin dashboard with subscriber management.
- Auto-dictionary with audio and synonyms is unique at this price in Vietnam.

SUMMARY

One App. Every Industry. Every Word.

The four things no single competitor matches — all in one product.

True SM-2 Spaced Repetition

Full algorithm — ease factor, interval, review count — not a simplified version.

CSV Bulk Import

Header detection, duplicate skipping, preview, and batch tagging at scale.

VietQR + Paddle Payments

Local Techcombank transfer for Vietnam users. International cards via Paddle.

Full Admin B2B Console

Password-protected dashboard with subscriber management and manual activation.